Pre-submission inquiry to Genome Medicine

Date: 8 November 2026

Subject: Pre-submission inquiry: analysis-unit dependence and effect-aware calibration in human kidney single-cell studies

Dear Editors,

We are considering submitting our manuscript, "Analysis-unit dependence and effect-aware calibration in human kidney thick ascending limb single-cell studies", as a Research Article to Genome Medicine.

Single-cell disease studies often describe a cell state as disease-associated without specifying whether the signal is a between-donor contrast, a tissue neighborhood association, or a within-donor cell-state correlation. We developed a unit-aware framework and evaluated it in 106,851 human kidney thick ascending limb cells from 223 donors. In aTAL2, donor-level and neighborhood-expression directions were positive while the within-donor association was negative. A prespecified 400-gene panel showed donor-within reversal in 19.0% of genes versus 12.7% of expression-matched random sets. A post hoc effect-aware pairing null gave 20.8%, showing that the initial excess reflected the selected effect-size distribution rather than broad negative coupling between units. A smaller neighborhood-versus-within excess remained threshold sensitive. The same expression-matched versus effect-aware divergence recurred in two external cohorts. A simulation benchmark distinguished true opposing effects from concordant and null scenarios.

Two non-KPMP cohorts provided directional support for donor-level PAPPA2 disease elevation, although exact subtype replication and the external within-donor effect were not obtained. Kidney tubulointerstitial eQTL, PTWAS, and snATAC analyses are reported as regulatory context and boundary evidence rather than causal proof. In GSE175759, an independent cohort of 87 primary kidney-disease samples, the frozen 11-gene TAL-injury score was associated with lower eGFR after diagnosis adjustment (beta = -8.71 eGFR points per SD; 95% CI -13.79 to -3.63; P = 0.0012). The disease-only estimate remained negative, whereas the IgA-only subgroup was underpowered. We present this as one independent clinical-association cohort, not as stable multicohort validation or clinical utility. The corrected analysis archive is available at https://doi.org/10.5281/zenodo.23811766.

We believe this work may fit Genome Medicine because it addresses a general measurement problem in genomic medicine and provides a reusable framework for cell-state studies across diseases. We are not claiming exact external subtype replication, experimental validation, or a clinical biomarker.

Before submitting, may we ask whether you consider this framework-focused study, with partial external support but without exact external subtype replication, suitable for the journal's scope and requirements, and whether it would be considered for external peer review?

We would be grateful for your guidance.

Yours sincerely,

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